

Detection Rates of GRB Afterglows with a Luminosity Distribution

An analytic derivation, continuing the bachelor's project and the project notes

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Summary

The bachelor's project (“Detection Rates of Gamma-Ray Burst Afterglows in Optical Surveys: An Analytic Approach”, referred to below as *the thesis*) and the project notes derive the rate $R_{\geq i}$ at which a survey with limiting flux F_{lim} and cadence t_{cad} detects afterglows at least i times, under the assumption that *all afterglows are identical* and only their distance D and viewing angle θ_{obs} vary. Here we relax that assumption in the simplest possible way: the afterglows keep a common light-curve *shape*, but their *luminosity* is drawn from a power-law distribution. We

1. summarise the single-luminosity results that are needed (Section 1),
2. introduce the luminosity distribution and work out how every quantity of the model scales with luminosity (Section 2),
3. write the full detection-rate integral (Section 3),
4. solve it analytically, case by case (Section 4), and
5. read off the detected-luminosity distribution and collect the final results (Sections 5–6).

The main outcome is that the detection rate of a population is *not* the single-luminosity rate at some typical luminosity: the rate of a single burst rises as $L^{3/2}$ until its detection horizon hits a wall (the cadence horizon reaching the edge of the Euclidean volume, or the whole beaming cone becoming visible), and saturates beyond it. Integrated over a power law $\propto L^\alpha$, this produces three regimes – dominated by the faint end, by the “knee” luminosity at the wall, or by the bright end – with different scalings of the rate with F_{lim} and t_{cad} than in the single-luminosity model.

Notation

All symbols are those of the thesis (equation numbers in square brackets refer to it). Only five new symbols are introduced, all in Section 2: the luminosity L , its distribution $\phi(L)$ with index α and bounds $L_{\text{min}}, L_{\text{max}}$, and the threshold luminosities $L_{\text{dec}}, L_{\text{j}}, L_{\text{nr}}, L_{\text{i}}$.

$q \equiv \theta_{\text{obs}}/\theta_j$, $\tilde{q} \equiv q - 1$	normalised viewing angle; θ_j = jet half-opening angle
$q_{\text{dec}} = 1 + (\Gamma_0 \theta_j)^{-1}$, $q_j = 2$, $q_{\text{nr}} = \sqrt{2}/\theta_j$	angles at which the off-axis light curve joins the on-axis one at t_{dec} and at t_j ; q_{nr} corresponds to $\theta_{\text{obs}} = \sqrt{2}$ [29–31]
t_{dec} , $t_j = t_{\text{dec}}(\Gamma_0 \theta_j)^{8/3}$, $\tilde{t}_j \equiv t_j/t_{\text{dec}}$	deceleration time, jet-break time; note $\tilde{q}_{\text{dec}} = \tilde{t}_j^{-3/8}$
$\tilde{F}_\nu(t) \equiv F_\nu(t)/F_\nu(t_{\text{dec}})$	normalised on-axis light curve (PLS G) [21]
$t_p(q)$, $\tilde{F}_p(q) \equiv F_p(q)/F_\nu(t_{\text{dec}})$	peak time and normalised peak flux of an off-axis observer [32, 33]
D , D_{Euc} , $\tilde{D} \equiv D/D_{\text{Euc}}$	distance; radius of the Euclidean volume; normalised distance
$\mathcal{R}$, $R_{\text{int}} = \frac{4}{3}\pi\mathcal{R}D_{\text{Euc}}^3$	GRB rate density; total GRB rate inside D_{Euc} [35, 36]
F_{lim} , $\tilde{F}_{\text{lim}} \equiv F_{\text{lim}}/F_\nu(t_{\text{dec}}, D_{\text{Euc}})$	limiting flux of the survey; normalised limiting flux [38]
$D_{\text{max}}(q)$, q_{Euc}	maximal detectable distance at angle q ; angle at which $D_{\text{max}} = D_{\text{Euc}}$ [38, 40]
t_{cad} , i , τ_{det} , $P_{\geq i}$	cadence; required number of detections; $\Delta t_{\text{det}}/t_{\text{cad}}$; probability of $\geq i$ detections [54, 60]
D_i , q_i	distance at which $\tau_{\text{det}} = i$; angle at which $D_{\text{max}}(q_i) = D_i$ [63, 64]
R_{det} , $R_{\geq i}$, $R(q, D) \equiv \frac{1}{2}\theta_j^2 q^2 \tilde{D}^3 R_{\text{int}}$	detection rates; rate contributed by the box $[0, q] \times [0, D]$ (thesis p. 24)
$L \equiv L_\nu(t_{\text{dec}})$	on-axis spectral luminosity of a burst at t_{dec} – the normalisation of [34]
$\phi(L) \propto L^\alpha$ on $[L_{\text{min}}, L_{\text{max}}]$	luminosity distribution (a probability density in L)
L_{dec} , L_j , L_{nr} , L_i	luminosities at which $q_{\text{Euc}}(L) = q_{\text{dec}}$, q_j , q_{nr} , q_i
$\langle R_{\geq i} \rangle$	detection rate averaged over the luminosity distribution

The index of the luminosity distribution is called α as is customary for GRB luminosity functions; it is the only α in this document (the temporal decay indices of thesis Table 2 are always written out as $-3(p-1)/4$ and $-p$). The detection count is called i throughout, following the later sections of the project notes (the thesis uses k); consequently i is never used as a generic index here.

1 Summary of the single-luminosity results

1.1 Population, light curve and the detection integral

The population model of thesis §3.1 is: a spherical Euclidean universe of radius D_{Euc} ; GRBs distributed uniformly in volume with rate density $\mathcal{R}$ and uniformly in $\cos\theta_{\text{obs}}$; and *identical internal parameters* for all afterglows, so that the only free parameters of an event are D and θ_{obs} . The total GRB rate inside the volume is

$$R_{\text{int}} = \mathcal{R} \int_{V_{\text{Euc}}} dV = \frac{4}{3}\pi\mathcal{R}D_{\text{Euc}}^3. \quad (\text{thesis 36})$$

An observer at $q = \theta_{\text{obs}}/\theta_j$ sees the on-axis light curve only from the joining time $t_p(q)$ onwards,

$$t_p(q) = \begin{cases} t_{\text{dec}}, & q < q_{\text{dec}}, \\ t_j \tilde{q}^{8/3}, & q_{\text{dec}} < q < q_j \text{ (phase II)}, \\ t_j \tilde{q}^2, & q_j < q < q_{\text{nr}} \text{ (phase III)}, \end{cases} \quad (\text{thesis 32})$$

so the peak flux received is the on-axis flux at that time, $F_p(q) = F_\nu(t_p(q))$. With the PLS G on-axis light curve normalised at t_{dec} ,

$$\tilde{F}_\nu(t) \equiv \frac{F_\nu(t)}{F_\nu(t_{\text{dec}})} = \begin{cases} (t/t_{\text{dec}})^{-3(p-1)/4}, & t_{\text{dec}} < t < t_j \text{ (phase II)}, \\ \tilde{t}_j^{-3(p-1)/4} (t/t_j)^{-p}, & t_j < t < t_{\text{nr}} \text{ (phase III)}, \end{cases} \quad (\text{thesis 21})$$

the normalised peak flux becomes

$$\tilde{F}_p(q) \equiv \frac{F_p(q)}{F_\nu(t_{\text{dec}})} = \tilde{F}_\nu(t_p(q)) = \begin{cases} 1, & q < q_{\text{dec}}, \\ (\tilde{q}/\tilde{q}_{\text{dec}})^{-2(p-1)} = \tilde{t}_j^{-3(p-1)/4} \tilde{q}^{-2(p-1)}, & q_{\text{dec}} < q < q_j, \\ \tilde{q}_{\text{dec}}^{-2} (\tilde{q}/\tilde{q}_{\text{dec}})^{-2p} = \tilde{t}_j^{-3(p-1)/4} \tilde{q}^{-2p}, & q_j < q < q_{\text{nr}}, \end{cases} \quad (\text{thesis 33})$$

where the two forms on each line are equal because $\tilde{q}_{\text{dec}} = \tilde{t}_j^{-3/8}$. $\tilde{F}_p(q)$ is a decreasing function of q : the further off-axis, the later the light curve is joined and the fainter its peak.

The flux received from a burst at distance D and angle q is then

$$F_p(q, D) = \frac{L_\nu(t_{\text{dec}}) \tilde{F}_p(q)}{4\pi D^2}, \quad (\text{thesis 34})$$

where $L_\nu(t_{\text{dec}}) = 4\pi D^2 F_\nu(t_{\text{dec}}, D)$ is the on-axis spectral luminosity at the deceleration time; it is this single number that fixes the brightness of the whole (identical) population, and it is the quantity we will later distribute. A survey detects the burst if $F_p(q, D) > F_{\text{lim}}$, so the detection rate is

$$R_{\text{det}} = 4\pi\theta_j^2 \mathcal{R} \int_0^{q_{\text{nr}}} dq q \int_0^{D_{\text{Euc}}} dD D^2 \Theta(F_p(q, D) - F_{\text{lim}}), \quad (\text{thesis 37})$$

where the small-angle approximation $d \cos \theta_{\text{obs}} \approx \theta_j^2 q dq$ was used, and $q_{\text{nr}} = \sqrt{2}/\theta_j$ is defined so that $\frac{1}{2}\theta_j^2 q_{\text{nr}}^2 = 1$ exactly (the beaming factor $f_b = \theta_j^2/2$ is thus built in: $R_{\text{det}} \rightarrow R_{\text{int}}$ when everything is detected). The distance out to which a burst at angle q can be seen follows from $F_p(q, D_{\text{max}}) = F_{\text{lim}}$,

$$\tilde{D}_{\text{max}}(q) \equiv \frac{D_{\text{max}}(q)}{D_{\text{Euc}}} = \tilde{F}_{\text{lim}}^{-1/2} \tilde{F}_p(q)^{1/2}, \quad \tilde{F}_{\text{lim}} \equiv \frac{F_{\text{lim}}}{F_\nu(t_{\text{dec}}, D_{\text{Euc}})} = \frac{4\pi D_{\text{Euc}}^2 F_{\text{lim}}}{L_\nu(t_{\text{dec}})}, \quad (\text{thesis 38})$$

and performing the D integral in [37] gives

$$R_{\text{det}} = \theta_j^2 R_{\text{int}} \int_0^{q_{\text{nr}}} dq q \min(\tilde{D}_{\text{max}}^3(q), 1). \quad (\text{thesis 39})$$

The min expresses the two possible walls of the detection region: the flux horizon $D_{\text{max}}(q)$ or the edge of the volume D_{Euc} .

1.2 The Euclidean angle q_{Euc} and the most-contributing point

The angle at which the flux horizon reaches the edge of the volume, $D_{\text{max}}(q_{\text{Euc}}) = D_{\text{Euc}}$, i.e. $\tilde{F}_p(q_{\text{Euc}}) = \tilde{F}_{\text{lim}}$, is

$$\tilde{q}_{\text{Euc}} = \begin{cases} (\Gamma_0 \theta_j)^{-1} \tilde{F}_{\text{lim}}^{-1/(2(p-1))}, & q_{\text{dec}} < q_{\text{Euc}} < q_j, \\ (\Gamma_0 \theta_j)^{-(p-1)/p} \tilde{F}_{\text{lim}}^{-1/(2p)}, & q_j < q_{\text{Euc}} < q_{\text{nr}}. \end{cases} \quad (\text{thesis 40})$$

For angles $q < q_{\text{Euc}}$ every burst inside the volume is detected; for $q > q_{\text{Euc}}$ only those closer than $D_{\text{max}}(q) < D_{\text{Euc}}$. The limiting fluxes at which q_{Euc} coincides with the three special angles separate the four ranges of the thesis,

$$\tilde{F}_{\text{lim,dec}} = \tilde{F}_p(q_{\text{dec}}) = 1, \quad \tilde{F}_{\text{lim,j}} = \tilde{F}_p(q_j) = \tilde{q}_{\text{dec}}^{2(p-1)}, \quad \tilde{F}_{\text{lim,nr}} = \tilde{F}_p(q_{\text{nr}}) = \tilde{q}_{\text{dec}}^{-2} (\tilde{q}_{\text{nr}}/\tilde{q}_{\text{dec}})^{-2p}, \quad (\text{thesis 45})$$

range I being $\tilde{F}_{\text{lim}} > 1$ (even an on-axis burst at D_{Euc} is too faint), range II $\tilde{F}_{\text{lim,j}} < \tilde{F}_{\text{lim}} < 1$, range III $\tilde{F}_{\text{lim,nr}} < \tilde{F}_{\text{lim}} < \tilde{F}_{\text{lim,j}}$, and range IV $\tilde{F}_{\text{lim}} < \tilde{F}_{\text{lim,nr}}$ (everything is detected).¹

¹Thesis Eq. 45 as printed shows $\tilde{F}_{\text{lim,j}} = \tilde{q}_{\text{dec}}^{-2(p-1)}$; the positive exponent is the one consistent with Eq. [33] and with the value $10^{-4.5}$ quoted in thesis Fig. 11 for $\tilde{q}_{\text{dec}} = 10^{-1.5}$, $p = 2.5$.

The exact integral [39] is elementary but cumbersome (thesis Eqs. 41–44). The thesis instead uses the density of detected events per logarithmic interval,

$$\frac{dR_{\text{det}}}{d \log D d \log q} = 4\pi\theta_j^2 \mathcal{R} q^2 D^3 = 6R_{\text{int}} \left(\frac{q}{q_{\text{nr}}}\right)^2 \tilde{D}^3, \quad (\text{thesis 46})$$

to locate the *most contributing point*: along the flux horizon $D_{\text{max}}(q) \propto \tilde{q}^{-(p-1)}$ or $\tilde{q}^{-p}$ falls faster than $q^{-2/3}$, so $q^2 D_{\text{max}}^3(q)$ decreases with q and the detections are dominated by the smallest angle on the horizon, i.e. by the corner where $D_{\text{max}}(q)$ meets whatever caps the distance (D_{Euc} here; the cadence distance D_i below). The rate is then approximated by the rate of the box $[0, q_\star] \times [0, D_\star]$ below and to the left of that corner,

$$R(q, D) \equiv \theta_j^2 R_{\text{int}} \int_0^q dq' q' \tilde{D}^3 = \frac{1}{2} \theta_j^2 q^2 \tilde{D}^3 R_{\text{int}}, \quad (1)$$

evaluated at $(q_\star, D_\star)$ (thesis p. 24; project notes Eq. 29). This gives thesis Eq. 52,

$$R_{\text{det}} \approx \frac{1}{2} \theta_j^2 R_{\text{int}} \begin{cases} q_{\text{dec}}^2 \tilde{F}_{\text{lim}}^{-3/2}, & 1 < \tilde{F}_{\text{lim}} \text{ (range I)}, \\ q_{\text{Euc}}^2 = \left((\Gamma_0 \theta_j)^{-1} \tilde{F}_{\text{lim}}^{-1/(2(p-1))} + 1 \right)^2, & \tilde{F}_{\text{lim,j}} < \tilde{F}_{\text{lim}} < 1 \text{ (range II)}, \\ q_{\text{Euc}}^2 = \left((\Gamma_0 \theta_j)^{-(p-1)/p} \tilde{F}_{\text{lim}}^{-1/(2p)} + 1 \right)^2, & \tilde{F}_{\text{lim,nr}} < \tilde{F}_{\text{lim}} < \tilde{F}_{\text{lim,j}} \text{ (range III)}, \\ q_{\text{nr}}^2 = 2\theta_j^{-2}, & \tilde{F}_{\text{lim}} < \tilde{F}_{\text{lim,nr}} \text{ (range IV)}, \end{cases} \quad (\text{thesis 52})$$

The corners are $(q_{\text{dec}}, D_{\text{dec}})$ in range I, with the on-axis horizon $\tilde{D}_{\text{dec}} = \tilde{F}_{\text{lim}}^{-1/2}$; $(q_{\text{Euc}}, D_{\text{Euc}})$ in ranges II and III; and $(q_{\text{nr}}, D_{\text{Euc}})$ in range IV, where everything is detected. Thesis Fig. 11 shows that this approximation reproduces the exact integral to within a factor of order unity everywhere (the worst case being a factor 4 near $\tilde{F}_{\text{lim,j}}$).

1.3 Cadence: the rate of $\geq i$ detections

A survey observes the same field every t_{cad} . The number of detections of an event is set by the ratio of the time it stays above the limiting flux to the cadence,

$$\tau_{\text{det}} \equiv \frac{\Delta t_{\text{det}}}{t_{\text{cad}}} \approx \frac{t_+(D)}{t_{\text{cad}}}, \quad F_\nu(t_+, D) = F_{\text{lim}} \iff \tilde{F}_\nu(t_+) = \tilde{D}^2 \tilde{F}_{\text{lim}}, \quad (\text{thesis 54, 55})$$

where t_+ is the time at which the on-axis light curve at distance D fades below F_{lim} (the approximation $\Delta t_{\text{det}} \approx t_+$ drops the joining time $t_- \approx t_p(q)$). The probability of at least i detections is

$$P_{\geq i}(\tau_{\text{det}}) = \begin{cases} 0, & \tau_{\text{det}} < i - 1, \\ \tau_{\text{det}} - (i - 1), & i - 1 \leq \tau_{\text{det}} < i, \\ 1, & i \leq \tau_{\text{det}}, \end{cases} \quad (\text{thesis 60})$$

and the rate of events detected at least i times is

$$R_{\geq i} = 4\pi\theta_j^2 \mathcal{R} \int_0^{q_{\text{nr}}} dq q \int_0^{D_{\text{Euc}}} dD D^2 \Theta(F_p(q, D) - F_{\text{lim}}) P_{\geq i}(\tau_{\text{det}}(D)). \quad (\text{thesis 58})$$

(The factor $\Theta(F_p - F_{\text{lim}})$ is implicit in the thesis through $\Delta t_{\text{det}} > 0$; it is written out here because the approximation $\Delta t_{\text{det}} \approx t_+$ no longer enforces it.) For $i \geq 2$ the thesis (§3.2.3) replaces the ramp by a step, $P_{\geq i} \approx \Theta(\tau_{\text{det}} - i)$, because the contribution of the ramp is dominated by its upper end; this is accurate to a few per cent. The step selects $D < D_i$, where D_i is the distance at which the on-axis light curve stays above the limit for exactly i cadences, $t_+(D_i) = i t_{\text{cad}}$:

$$\tilde{D}_i^2 = \frac{\tilde{F}_\nu(i t_{\text{cad}})}{\tilde{F}_{\text{lim}}} \implies \tilde{D}_i = \tilde{F}_{\text{lim}}^{-1/2} \begin{cases} (i t_{\text{cad}}/t_{\text{dec}})^{-3(p-1)/8}, & i t_{\text{cad}} < t_{\text{j}}, \\ \tilde{t}_{\text{j}}^{-3(p-1)/8} (i t_{\text{cad}}/t_{\text{j}})^{-p/2}, & i t_{\text{cad}} > t_{\text{j}}. \end{cases} \quad (\text{thesis 63})$$

The corresponding angle q_i is where the flux horizon meets this cap, $D_{\max}(q_i) = D_i$, i.e. $\tilde{F}_p(q_i) = \tilde{F}_\nu(i t_{\text{cad}})$, i.e. $t_p(q_i) = i t_{\text{cad}}$:

$$\tilde{q}_i = \begin{cases} (i t_{\text{cad}}/t_j)^{3/8}, & i t_{\text{cad}} < t_j, \\ (i t_{\text{cad}}/t_j)^{1/2}, & i t_{\text{cad}} > t_j. \end{cases} \quad (\text{thesis 64})$$

Bursts at $q < q_i$ have $D_{\max}(q) > D_i$, so for them the cadence cap D_i binds; bursts at $q > q_i$ peak later than $i t_{\text{cad}}$ and are limited by their own flux horizon. Altogether the detection region is

$$R_{\geq i} = \theta_j^2 R_{\text{int}} \int_0^{q_{\text{nr}}} dq q \min(\tilde{D}_{\max}(q), \tilde{D}_i, 1)^3, \quad (2)$$

the single-luminosity analogue of [39] with the cadence cap added. Which cap binds at the corner depends on whether D_i is smaller or larger than D_{Euc} ; since $D_{\max}$ decreases with q , $D_i < D_{\text{Euc}} \Leftrightarrow q_i > q_{\text{Euc}}$. The limiting flux at which the two coincide, $q_{\text{Euc}}(F_{\text{lim},i}) = q_i$, is

$$\tilde{F}_{\text{lim},i} = \tilde{F}_p(q_i) = \tilde{F}_\nu(i t_{\text{cad}}) = \begin{cases} (i t_{\text{cad}}/t_{\text{dec}})^{-3(p-1)/4}, & i t_{\text{cad}} < t_j, \\ \tilde{t}_j^{-3(p-1)/4} (i t_{\text{cad}}/t_j)^{-p} = \tilde{t}_j^{(p+3)/4} (i t_{\text{cad}}/t_{\text{dec}})^{-p}, & i t_{\text{cad}} > t_j, \end{cases} \quad (\text{thesis 69})$$

which is simply the on-axis light curve evaluated at $t = i t_{\text{cad}}$.²

Applying the most-contributing-point argument with the additional cap gives the regions listed in Table 1 (project notes Table 17; thesis Eqs. 73–75),³ and the compact result

$$R_{\geq i} = \begin{cases} R(\min(q_i, q_{\text{nr}}), D_i), & \tilde{F}_{\text{lim},i} < \tilde{F}_{\text{lim}} \quad (q_{\text{Euc}} < q_i, \text{ cadence-limited}), \\ R(\min(q_{\text{Euc}}, q_{\text{nr}}), D_{\text{Euc}}) = R_{\text{det}}, & \tilde{F}_{\text{lim}} < \tilde{F}_{\text{lim},i} \quad (q_i < q_{\text{Euc}}, \text{ Euclidean-limited}). \end{cases} \quad (\text{thesis 75})$$

Writing out the cadence-limited branch with (thesis 63), in phase III ($i t_{\text{cad}} > t_j$),

$$R(q_i, D_i) = \frac{1}{2} \theta_j^2 q_i^2 \tilde{D}_i^3 R_{\text{int}} = \frac{1}{2} \theta_j^2 (\Gamma_0 \theta_j)^{-3(p-1)} \left(\frac{i t_{\text{cad}}}{t_j} \right)^{-3p/2} \left(\left(\frac{i t_{\text{cad}}}{t_j} \right)^{1/2} + 1 \right)^2 \tilde{F}_{\text{lim}}^{-3/2} R_{\text{int}}, \quad (\text{thesis 73})$$

where $\tilde{t}_j^{-9(p-1)/8} = (\Gamma_0 \theta_j)^{-3(p-1)}$ was used. The steep $\tilde{F}_{\text{lim}}^{-3/2}$ dependence is the volume of the cadence-limited sphere, $D_i^3 \propto F_{\text{lim}}^{-3/2}$.

	$q_{\text{Euc}} < q_i$ (cadence-limited)	$q_i < q_{\text{Euc}}$ (Euclidean-limited)
$q_{\text{dec}} < \max(q_i, q_{\text{Euc}}) < q_j$	(q_i, D_i)	$(q_{\text{Euc}}, D_{\text{Euc}})$
$q_j < \max(q_i, q_{\text{Euc}}) < q_{\text{nr}}$	(q_i, D_i)	$(q_{\text{Euc}}, D_{\text{Euc}})$
$q_{\text{nr}} < \max(q_i, q_{\text{Euc}})$	(q_{nr}, D_i)	$(q_{\text{nr}}, D_{\text{Euc}})$

Table 1: Most contributing point $(q_\star, D_\star)$ for $i \geq 2$ detections, assuming $q_{\text{dec}} < q_i$. In the left column $D_i < D_{\text{Euc}}$, in the right column $D_{\text{Euc}} < D_i$. The rate is $R_{\geq i} \approx R(q_\star, D_\star) = \frac{1}{2} \theta_j^2 q_\star^2 \tilde{D}_\star^3 R_{\text{int}}$. In range I ($\tilde{F}_{\text{lim}} > 1$, no q_{Euc} exists) the corner is always (q_i, D_i) . The case $q_i < q_{\text{dec}}$, whose corner would be $(q_{\text{dec}}, D_{\text{dec}})$, requires $i t_{\text{cad}} < t_{\text{dec}} \approx 20$ s and plays no role for optical surveys; it is excluded from now on.

²The second line of thesis Eq. 69 is printed with $\tilde{t}_j^{3(p-1)/4}$; the exponent $(p+3)/4$ is the one consistent with Eqs. [33], [63] and [64] (the printed form is off by exactly $\tilde{t}_j^{(3-p)/2}$).

³In project notes Table 17 the two column headings are interchanged relative to thesis Eqs. 74–75. The cadence-limited point (q_i, D_i) applies when $q_{\text{Euc}} < q_i$, because then $D_i < D_{\text{Euc}}$.

1.4 Remark on real survey parameters

In the project notes the survey is characterised by its exposure time, number of pointings, sky coverage and live fraction, and the limiting flux follows from $F_{\text{lim}} = A (t_{\text{exp}}/t_{\text{exp},0})^{-1/2}$ with $t_{\text{exp}} = f_{\text{live}} t_{\text{cad}}/N_{\text{exp}}$; the rate of the surveyed sky fraction is $f_{\Omega} R_{\geq i}$ with $f_{\Omega} = \Omega_{\text{srv}}/4\pi$ (thesis Eq. 78). None of this interacts with the luminosity distribution: the survey enters the rate only through the three numbers $(F_{\text{lim}}, t_{\text{cad}}, i)$ and the overall factor f_{Ω} . Everything below is therefore derived at fixed $(F_{\text{lim}}, t_{\text{cad}}, i)$ for the full sky, exactly as in thesis §3.2.

2 Introducing the luminosity distribution

2.1 Motivation

Observed afterglows at a fixed epoch span several orders of magnitude in luminosity, whereas the model of Section 1 assigns all of them the single value $L_{\nu}(t_{\text{dec}})$. One might hope to absorb the spread into a suitably chosen “typical” luminosity, but the detection rate is far from linear in L : a fainter burst loses detection *volume*, since every horizon distance scales as $D \propto L^{1/2}$ and the volume as $L^{3/2}$ (halving L removes about 65% of the detections), while a brighter burst gains little once its horizon reaches a wall – the edge D_{Euc} of the volume in the cadence-limited regions, or the full beaming cone q_{nr} . The rate of a single burst is thus a *rising-then-saturating* function of its luminosity, and the average of such a function over a broad distribution is not its value at any average luminosity. We now make this quantitative.

2.2 Definition and assumptions

The luminosity variable. We label each burst by its on-axis spectral luminosity at the deceleration time,

$$L \equiv L_{\nu}(t_{\text{dec}}) = 4\pi D^2 F_{\nu}(t_{\text{dec}}, D), \quad (3)$$

which is exactly the normalisation of the peak flux in thesis Eq. 34, $F_p(q, D) = L \tilde{F}_p(q)/(4\pi D^2)$.

Its physical content follows from thesis Eq. 24: $L \propto \bar{\epsilon}_e^{p-1} \epsilon_B^{(p+1)/4} n^{1/2} E_{\text{k,iso}}^{(p+3)/4} t_{\text{dec}}^{-3(p-1)/4}$.

The “gray” assumption. All bursts share the same light-curve *shape*, i.e. the same t_{dec} , t_j , p , Γ_0 and θ_j , and therefore the same $\tilde{F}_{\nu}(t)$, $\tilde{F}_p(q)$, $t_p(q)$ and the same angles $q_{\text{dec}}, q_j, q_{\text{nr}}$; only the normalisation L varies from burst to burst. Physically this is the case for a spread in the microphysical parameters ϵ_e and ϵ_B , which enter only the normalisation. A spread in $E_{\text{k,iso}}$ or n would also move $t_{\text{dec}} \propto E_{\text{k,iso}}^{1/3} n^{-1/3}$ and t_j , changing the shape; we neglect this (it moves the comparison points by factors of order $(\Delta E)^{1/3}$, well below the accuracy of the Euclidean framework).

The distribution. We take a power law between a minimum and a maximum luminosity,

$$\phi(L) dL = \frac{(\alpha + 1) L^{\alpha}}{L_{\text{max}}^{\alpha+1} - L_{\text{min}}^{\alpha+1}} dL, \quad L_{\text{min}} \leq L \leq L_{\text{max}}, \quad \int_{L_{\text{min}}}^{L_{\text{max}}} \phi(L) dL = 1 \quad (4)$$

(for $\alpha = -1$ the prefactor is replaced by its limit, $\phi(L) = 1/(L \ln(L_{\text{max}}/L_{\text{min}}))$). Because ϕ is normalised to one, R_{int} keeps its meaning as the total GRB rate inside D_{Euc} : the distribution only redistributes brightness among the same bursts. Typical GRB luminosity functions have α between -1 and -3 ; the median luminosity of the population is

$$L_{\text{med,pop}}^{\alpha+1} = \frac{1}{2} (L_{\text{min}}^{\alpha+1} + L_{\text{max}}^{\alpha+1}), \quad (5)$$

which for $\alpha < -1$ and $L_{\text{max}} \gg L_{\text{min}}$ sits at $L_{\text{med,pop}} \approx 2^{1/|\alpha+1|} L_{\text{min}}$, close to the faint end.

Relation to the observationally quoted luminosity. Afterglow luminosities are usually quoted as νL_{ν} at one day. Under the gray assumption this is a fixed multiple of L ,

$$\nu L_{\nu}(1 \text{ d}) = \nu \tilde{F}_{\nu}(1 \text{ d}) L, \quad \tilde{F}_{\nu}(1 \text{ d}) = \left(\frac{1 \text{ d}}{t_{\text{dec}}} \right)^{-3(p-1)/4} \quad (t_j > 1 \text{ d}), \quad (6)$$

so a power law in L is the same power law, with the same index α , in $\nu L_\nu(1\text{ d})$; only the bounds are rescaled. For the thesis parameters ($\nu = 5 \times 10^{14}$ Hz, $t_{\text{dec}} = 19.4$ s, $p = 2.5$) the factor is $\nu \tilde{F}_\nu(1\text{ d}) \approx 3.9 \times 10^{10}$ Hz, and the fiducial burst, $F_\nu(t_{\text{dec}}, D_{\text{Euc}}) = 2.3$ Jy at $D_{\text{Euc}} = 1.63 \times 10^{28}$ cm, has $L = 7.7 \times 10^{34}$ erg s $^{-1}$ Hz $^{-1}$ and $\nu L_\nu(1\text{ d}) \approx 3 \times 10^{45}$ erg s $^{-1}$.

2.3 How the model scales with luminosity

The central observation is that luminosity enters Section 1 in exactly one place: the normalised limiting flux,

$$\tilde{F}_{\text{lim}} = \frac{4\pi D_{\text{Euc}}^2 F_{\text{lim}}}{L} \equiv \frac{L_{\text{dec}}}{L}, \quad L_{\text{dec}} \equiv 4\pi D_{\text{Euc}}^2 F_{\text{lim}}. \quad (7)$$

L_{dec} is the luminosity of a burst that, seen on-axis from the edge of the volume, is exactly at the flux limit at t_{dec} . It plays the role of $\tilde{F}_{\text{lim,dec}} = 1$ in [45]: a burst with $L < L_{\text{dec}}$ is in range I of the survey, a burst with $L > L_{\text{dec}}$ in ranges II–IV. For a ZTF-like limit ($F_{\text{lim}} = 3 \times 10^{-5}$ Jy) and $D_{\text{Euc}} = 1.63 \times 10^{28}$ cm one finds $L_{\text{dec}} \approx 1.0 \times 10^{30}$ erg s $^{-1}$ Hz $^{-1}$, so the thesis' fiducial burst has $L/L_{\text{dec}} \approx 10^{4.9}$, i.e. $\tilde{F}_{\text{lim}} \approx 1.3 \times 10^{-5}$ as in thesis §3.3.

Substituting (7) into the single-luminosity formulas gives the scalings of every quantity with L :

Distances scale as $L^{1/2}$. From [38] and (thesis 63),

$$D_{\text{max}}(q; L) = \left(\frac{L \tilde{F}_p(q)}{4\pi F_{\text{lim}}} \right)^{1/2}, \quad \tilde{D}_{\text{max}}(q; L) = \left(\frac{L}{L_{\text{dec}}} \right)^{1/2} \tilde{F}_p(q)^{1/2}, \quad (8)$$

$$D_{\text{dec}}(L) = \left(\frac{L}{4\pi F_{\text{lim}}} \right)^{1/2}, \quad \tilde{D}_{\text{dec}}(L) = \left(\frac{L}{L_{\text{dec}}} \right)^{1/2}, \quad (9)$$

$$D_i(L) = \left(\frac{L \tilde{F}_\nu(i t_{\text{cad}})}{4\pi F_{\text{lim}}} \right)^{1/2}, \quad \tilde{D}_i(L) = \left(\frac{L}{L_{\text{dec}}} \right)^{1/2} \tilde{F}_\nu(i t_{\text{cad}})^{1/2} \equiv \left(\frac{L}{L_i} \right)^{1/2}. \quad (10)$$

Angles and times do not scale. q_{dec} , q_j , q_{nr} , t_{dec} , t_j , $t_p(q)$ are shape quantities, and so is the cadence angle q_i [64], which is fixed by $t_p(q_i) = i t_{\text{cad}}$ alone. The only angle that depends on L is the Euclidean angle q_{Euc} : from $\tilde{F}_p(q_{\text{Euc}}) = \tilde{F}_{\text{lim}} = L_{\text{dec}}/L$ and [40],

$$\tilde{q}_{\text{Euc}}(L) = \begin{cases} (\Gamma_0 \theta_j)^{-1} \left(\frac{L}{L_{\text{dec}}} \right)^{\frac{1}{2(p-1)}} = \left(\frac{L}{L_j} \right)^{\frac{1}{2(p-1)}}, & L_{\text{dec}} < L < L_j \quad (q_{\text{dec}} < q_{\text{Euc}} < q_j), \\ (\Gamma_0 \theta_j)^{-\frac{p-1}{p}} \left(\frac{L}{L_{\text{dec}}} \right)^{\frac{1}{2p}} = \left(\frac{L}{L_j} \right)^{\frac{1}{2p}}, & L_j < L < L_{\text{nr}} \quad (q_j < q_{\text{Euc}} < q_{\text{nr}}), \end{cases} \quad (11)$$

a slowly increasing function of L (exponents $1/3$ and $1/5$ for $p = 2.5$): a brighter burst is visible from D_{Euc} further off-axis, but only slightly.

The threshold luminosities. The second form in (11) uses the luminosities at which $q_{\text{Euc}}(L)$ passes the special angles. In general, define for any angle q_X the luminosity at which a burst at that angle is exactly at the flux limit when seen from D_{Euc} :

$$L_X \equiv \frac{4\pi D_{\text{Euc}}^2 F_{\text{lim}}}{\tilde{F}_p(q_X)} = \frac{L_{\text{dec}}}{\tilde{F}_p(q_X)} \iff q_{\text{Euc}}(L_X) = q_X. \quad (12)$$

These are the exact analogues of the limiting fluxes $\tilde{F}_{\text{lim},X}$ of [45] and [69]: $\tilde{F}_{\text{lim},X} = L_{\text{dec}}/L_X$. Explicitly, using (thesis 33),

$$L_j = L_{\text{dec}} \tilde{q}_{\text{dec}}^{-2(p-1)} = L_{\text{dec}} (\Gamma_0 \theta_j)^{2(p-1)}, \quad (13)$$

$$L_{\text{nr}} = L_{\text{dec}} \tilde{q}_{\text{dec}}^2 (\tilde{q}_{\text{nr}}/\tilde{q}_{\text{dec}})^{2p} = L_j \tilde{q}_{\text{nr}}^{2p}, \quad \tilde{q}_{\text{nr}} = \frac{\sqrt{2}}{\theta_j} - 1, \quad (14)$$

$$L_i = \frac{L_{\text{dec}}}{\tilde{F}_\nu(i t_{\text{cad}})} = \begin{cases} L_{\text{dec}} \left(\frac{i t_{\text{cad}}}{t_{\text{dec}}} \right)^{3(p-1)/4} = L_j \tilde{q}_i^{2(p-1)}, & i t_{\text{cad}} < t_j, \\ L_{\text{dec}} (\Gamma_0 \theta_j)^{2(p-1)} \left(\frac{i t_{\text{cad}}}{t_j} \right)^p = L_j \tilde{q}_i^{2p}, & i t_{\text{cad}} > t_j. \end{cases} \quad (15)$$

L_j is the luminosity needed to see, from D_{Euc} , a burst whose light curve is joined at the jet break; L_{nr} the luminosity needed to see the whole beaming cone from D_{Euc} ; and L_i the luminosity of a burst whose on-axis light curve, placed at D_{Euc} , drops to the flux limit exactly i cadences after the burst – equivalently, the luminosity at which the cadence distance D_i reaches the edge of the volume, $\tilde{D}_i(L_i) = 1$ by (10). Since $\tilde{F}_p$ decreases with q , the ordering of the thresholds is the ordering of the angles:

$$L_{\text{dec}} < L_j < L_{\text{nr}}, \quad \text{and} \quad L_i \leq L_{\text{dec}}, L_j, L_{\text{nr}} \iff q_i \leq q_{\text{dec}}, q_j, q_{\text{nr}}. \quad (16)$$

Most importantly, because $q_{\text{Euc}}(L)$ increases with L ,

$$q_{\text{Euc}}(L) < q_i \iff L < L_i \iff D_i(L) < D_{\text{Euc}} : \quad (17)$$

a burst fainter than L_i is cadence-limited, a burst brighter than L_i is Euclidean-limited. The regime of a burst is decided by its luminosity alone.

2.4 The single-burst rate as a function of luminosity

We can now rewrite thesis Eqs. 73–76 as a function of L at fixed survey. Inserting (10)–(17) into [75] and (1), for $q_{\text{dec}} < q_i$:

$$R_{\geq i}(L) = \frac{1}{2}\theta_j^2 R_{\text{int}} \times \begin{cases} \min(q_i, q_{\text{nr}})^2 \left(\frac{L}{L_i}\right)^{3/2}, & L < L_i, \\ \left(1 + \left(\frac{L}{L_j}\right)^{\frac{1}{2(p-1)}}\right)^2, & L_i < L < L_j \quad (\text{only if } q_i < q_j), \\ \left(1 + \left(\frac{L}{L_j}\right)^{\frac{1}{2p}}\right)^2, & \max(L_i, L_j) < L < L_{\text{nr}} \quad (\text{only if } q_i < q_{\text{nr}}), \\ q_{\text{nr}}^2 = \frac{2}{\theta_j^2}, & L > \max(L_i, L_{\text{nr}}). \end{cases} \quad (18)$$

The four lines are, in order: the cadence-limited regime, with corner (q_i, D_i) ; the Euclidean-limited regime, with corner $(q_{\text{Euc}}(L), D_{\text{Euc}})$, in phase II and in phase III; and the saturated regime, in which everything inside D_{Euc} is detected. In the thesis form, the cadence-limited line reads, for $i t_{\text{cad}} > t_j$,

$$R_{\geq i}(L) = \frac{1}{2}\theta_j^2 (\Gamma_0 \theta_j)^{-3(p-1)} \left(\frac{i t_{\text{cad}}}{t_j}\right)^{-3p/2} \left(\left(\frac{i t_{\text{cad}}}{t_j}\right)^{1/2} + 1\right)^2 \left(\frac{L}{L_{\text{dec}}}\right)^{3/2} R_{\text{int}}, \quad L < L_i, \quad (19)$$

i.e. thesis Eq. 73 with $\tilde{F}_{\text{lim}}^{-3/2} \rightarrow (L/L_{\text{dec}})^{3/2}$. The two forms agree because, by (15), $L_i^{-3/2} = L_{\text{dec}}^{-3/2} \tilde{F}_\nu(i t_{\text{cad}})^{3/2}$.

$R_{\geq i}(L)$ is continuous: at $L = L_i$ the first line gives $\min(q_i, q_{\text{nr}})^2$ and the appropriate Euclidean-limited line gives $q_{\text{Euc}}(L_i)^2 = q_i^2$ (or q_{nr}^2 if $q_i > q_{\text{nr}}$); at $L = L_j$ both phase branches give $(1 + 1)^2 = q_j^2$; at $L = L_{\text{nr}}$ the phase-III line gives $(1 + \tilde{q}_{\text{nr}})^2 = q_{\text{nr}}^2$. It rises steeply, as $L^{3/2}$, while the cadence horizon $D_i(L)$ grows inside the volume; once D_i reaches D_{Euc} (at L_i) further luminosity only widens the visible cone, $R \propto q_{\text{Euc}}(L)^2$, which grows as $L^{1/(p-1)}$ at most and as $L^{1/p}$ beyond the jet-break angle; and once the whole cone is visible (at L_{nr}) the rate saturates at R_{int} . Figure 1 shows this for the thesis parameters. Expressed in units of L_{dec} and R_{int} the curves are universal: the survey enters only through the position of $L_i/L_{\text{dec}} = 1/\tilde{F}_\nu(i t_{\text{cad}})$.

3 The full integral

3.1 Definition

Each burst is now characterised by three quantities, (q, D, L) , distributed independently: q and D as before (uniform in volume and in $\cos \theta_{\text{obs}}$), L according to $\phi(L)$. The rate of events detected

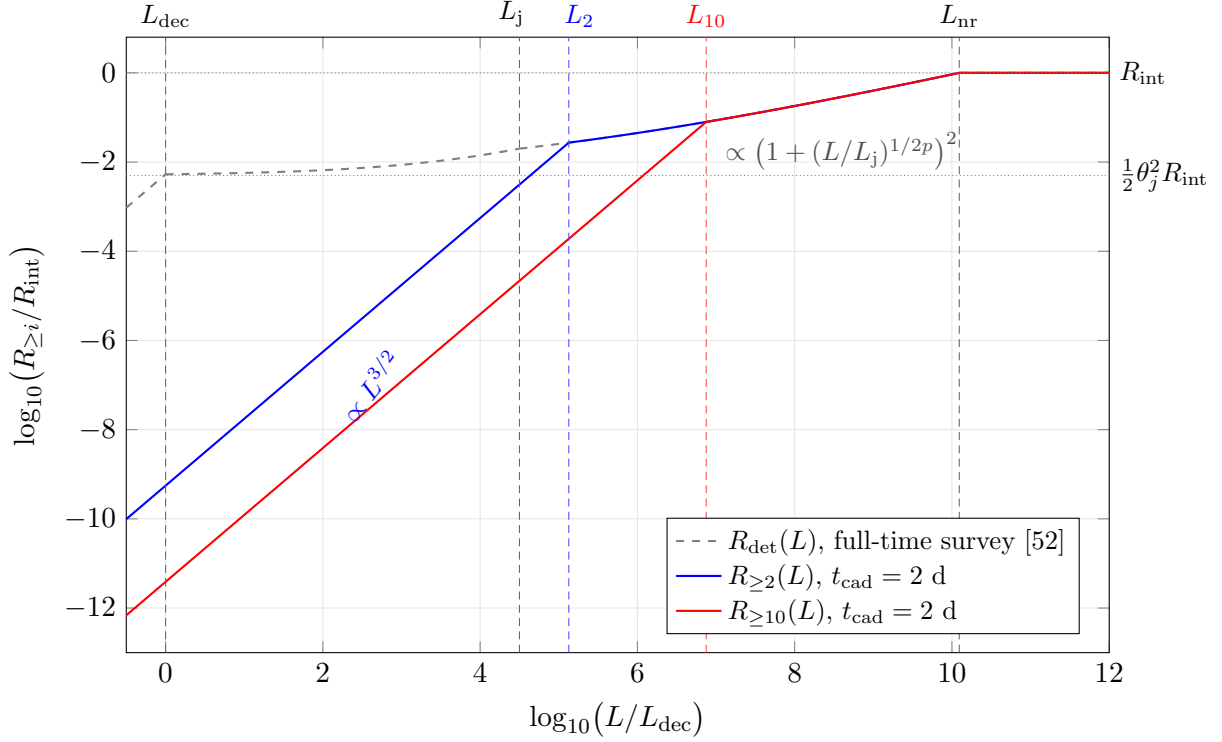


Figure 1: The single-burst detection rate as a function of its luminosity, Eq. (18), for the thesis parameters of Appendix B ($p = 2.5$, $\Gamma_0\theta_j = 10^{1.5}$, $\theta_j = 0.1$, $t_j = 10^4 t_{\text{dec}}$) and a cadence of two days, for $i = 2$ (blue) and $i = 10$ (red). The dashed grey line is the full-time survey rate, thesis Eq. 52, which the $\geq i$ rates join at L_i . Below L_i the rate is cadence-limited and rises as $L^{3/2}$; between L_i and L_{nr} it is Euclidean-limited and grows only as the visible cone widens; above L_{nr} every burst inside D_{Euc} is detected. In these units the figure is independent of F_{lim} and D_{Euc} .

at least i times is thesis Eq. 58 integrated over the luminosity distribution,

$$\langle R_{\geq i} \rangle = \int_{L_{\min}}^{L_{\max}} dL \phi(L) 4\pi\theta_j^2 \mathcal{R} \int_0^{q_{\text{nr}}} dq q \int_0^{D_{\text{Euc}}} dD D^2 \Theta\left(\frac{L \tilde{F}_p(q)}{4\pi D^2} - F_{\text{lim}}\right) \Theta(t_+(D; L) - i t_{\text{cad}}), \quad (20)$$

where, as in Section 1.3, the step approximation $P_{\geq i} \approx \Theta(\tau_{\text{det}} - i)$ for $i \geq 2$ has been used, and both the flux horizon and the cadence horizon now depend on L . Performing the D integral with (8) and (10),

$$\langle R_{\geq i} \rangle = \theta_j^2 R_{\text{int}} \int_{L_{\min}}^{L_{\max}} dL \phi(L) \int_0^{q_{\text{nr}}} dq q \min\left(\tilde{D}_{\text{max}}(q; L), \tilde{D}_i(L), 1\right)^3, \quad (21)$$

the luminosity average of (2).

3.2 The exact expression: integrating over luminosity first

Both L -dependent distances in (21) scale as $L^{1/2}$, so their minimum does too,

$$\min(\tilde{D}_{\text{max}}(q; L), \tilde{D}_i(L)) = \left(\frac{L}{L_{\text{dec}}}\right)^{1/2} \min(\tilde{F}_p(q), \tilde{F}_p(q_i))^{1/2} = \left(\frac{L}{L_{\text{dec}}}\right)^{1/2} \tilde{F}_p(\max(q, q_i))^{1/2}, \quad (22)$$

because $\tilde{F}_p$ decreases with q : at $q < q_i$ the cadence cap binds, at $q > q_i$ the flux horizon. With the threshold luminosity of the angle $\max(q, q_i)$ in the sense of (12), $L_{\text{Euc}}(q) \equiv L_{\text{dec}}/\tilde{F}_p(q)$, the

integrand is

$$\min\left(\tilde{D}_{\max}(q; L), \tilde{D}_i(L), 1\right)^3 = \begin{cases} \left(\frac{L}{L_{\text{Euc}}(\max(q, q_i))}\right)^{3/2}, & L < L_{\text{Euc}}(\max(q, q_i)), \\ 1, & L > L_{\text{Euc}}(\max(q, q_i)), \end{cases} \quad (23)$$

i.e. for a given angle, bursts fainter than $L_{\text{Euc}}(\max(q, q_i))$ are seen out to a horizon inside the volume ($\propto L^{3/2}$ of them), brighter ones throughout the volume. The luminosity integral is therefore elementary for every q :

$$\int_{L_{\min}}^{L_{\max}} dL \phi(L) \min\left(\tilde{D}_{\max}(q; L), \tilde{D}_i(L), 1\right)^3 = \frac{\alpha + 1}{L_{\max}^{\alpha+1} - L_{\min}^{\alpha+1}} \times \left[L_{\text{Euc}}(\max(q, q_i))^{-3/2} \frac{\min(L_{\max}, L_{\text{Euc}}(\max(q, q_i)))^{\alpha+5/2} - L_{\min}^{\alpha+5/2}}{\alpha + 5/2} + \frac{L_{\max}^{\alpha+1} - \max(L_{\min}, L_{\text{Euc}}(\max(q, q_i)))^{\alpha+1}}{\alpha + 1} \right], \quad (24)$$

where each of the two terms is to be set to zero when its upper limit falls below its lower one (the first when $L_{\text{Euc}}(\max(q, q_i)) < L_{\min}$, all bursts saturated; the second when $L_{\text{Euc}}(\max(q, q_i)) > L_{\max}$, none saturated). Inserting (24) into (21) leaves a single integral over q ,

$$\langle R_{\geq i} \rangle = \theta_j^2 R_{\text{int}} \int_0^{q_{\text{nr}}} dq q \times [\text{Eq. (24)}], \quad (25)$$

in which $L_{\text{Euc}}(\max(q, q_i))$ is a piecewise power law in $\tilde{q}$ through (thesis 33) (L_{dec} for $q < q_{\text{dec}}$; $L_j \tilde{q}^{2(p-1)}$ for $q_{\text{dec}} < q < q_j$; $L_j \tilde{q}^{2p}$ for $q > q_j$; constant L_i for all $q < q_i$). Equation (25) is the exact analogue of thesis Eqs. 41–44: every piece is integrable in closed form, but the result is a long sum of incomplete powers of the various $\tilde{q}$ boundaries (with $L_{\min}$ and $L_{\max}$ now cutting the q range at the angles where $L_{\text{Euc}}(q) = L_{\min}$ and $L_{\text{Euc}}(q) = L_{\max}$). It is the appropriate starting point for a numerical evaluation. As in the thesis, we proceed with the most-contributing-point approximation instead, which gives compact analytic results; in the cases we checked it reproduces (25) to within 5–25%, comparable to its accuracy for a single luminosity (thesis Fig. 11).

3.3 Reduction to one dimension

The most-contributing-point approximation replaces the q integral in (21), for each luminosity separately, by $R_{\geq i}(L)$ of Eq. (18). Since the regime of a burst is set by its luminosity alone (Eq. (17)), the average is a one-dimensional integral over a piecewise power law:

$$\langle R_{\geq i} \rangle = \int_{L_{\min}}^{L_{\max}} \phi(L) R_{\geq i}(L) dL \quad (26)$$

with ϕ from (4) and $R_{\geq i}(L)$ from (18). The rest of the derivation is the evaluation of this integral.

4 Analytic solution, case by case

4.1 The elementary integral

Every term in (26) is of the form $\int L^\alpha \cdot L^\beta dL$ with $\beta \in \{\frac{3}{2}, \frac{1}{2(p-1)}, \frac{1}{p-1}, \frac{1}{2p}, \frac{1}{p}, 0\}$, for which

$$\int_a^b L^{\alpha+\beta} dL = \begin{cases} \frac{b^{\alpha+\beta+1} - a^{\alpha+\beta+1}}{\alpha + \beta + 1}, & \alpha + \beta \neq -1, \\ \ln \frac{b}{a}, & \alpha + \beta = -1. \end{cases} \quad (27)$$

The logarithmic case occurs at isolated values of α (e.g. $\alpha = -5/2$ for the cadence-limited term, $\alpha = -1$ for the saturated term and for the normalisation of ϕ); they are collected in Appendix A and are the continuous limits of the general expressions.

4.2 The four segments

The integration range $[L_{\min}, L_{\max}]$ is cut by the thresholds L_i, L_j, L_{nr} into at most four segments, on each of which $R_{\geq i}(L)$ is one line of Eq. (18). Writing $[a, b]$ for the overlap of a segment with $[L_{\min}, L_{\max}]$ (a segment whose overlap is empty contributes nothing), the contributions are:

Cadence-limited segment, with $a = L_{\min}$ and $b = \min(L_{\max}, L_i)$. Here $\phi R \propto L^{\alpha+3/2}$:

$$\int_a^b \phi(L) R_{\geq i}(L) dL = \frac{1}{2} \theta_j^2 \min(q_i, q_{\text{nr}})^2 R_{\text{int}} \frac{\alpha + 1}{L_{\max}^{\alpha+1} - L_{\min}^{\alpha+1}} L_i^{-3/2} \frac{b^{\alpha+5/2} - a^{\alpha+5/2}}{\alpha + 5/2}. \quad (28)$$

It is the single-luminosity result $\frac{1}{2} \theta_j^2 \min(q_i, q_{\text{nr}})^2 \tilde{D}_i^3 R_{\text{int}}$ with $\tilde{D}_i^3 = (L/L_i)^{3/2}$ replaced by its average over the segment.

Euclidean-limited segment, phase II (exists only if $q_i < q_j$), with $a = \max(L_{\min}, L_i)$ and $b = \min(L_{\max}, L_j)$. Expanding the square in the second line of (18),

$$\left(1 + \left(\frac{L}{L_j}\right)^{\frac{1}{2(p-1)}}\right)^2 = 1 + 2\left(\frac{L}{L_j}\right)^{\frac{1}{2(p-1)}} + \left(\frac{L}{L_j}\right)^{\frac{1}{p-1}}, \quad (29)$$

gives three elementary integrals:

$$\begin{aligned} \int_a^b \phi(L) R_{\geq i}(L) dL = \frac{1}{2} \theta_j^2 R_{\text{int}} \frac{\alpha + 1}{L_{\max}^{\alpha+1} - L_{\min}^{\alpha+1}} & \left[\frac{b^{\alpha+1} - a^{\alpha+1}}{\alpha + 1} \right. \\ & \left. + 2 L_j^{-\frac{1}{2(p-1)}} \frac{b^{\alpha+1+\frac{1}{2(p-1)}} - a^{\alpha+1+\frac{1}{2(p-1)}}}{\alpha + 1 + \frac{1}{2(p-1)}} + L_j^{-\frac{1}{p-1}} \frac{b^{\alpha+1+\frac{1}{p-1}} - a^{\alpha+1+\frac{1}{p-1}}}{\alpha + 1 + \frac{1}{p-1}} \right]. \quad (30) \end{aligned}$$

Euclidean-limited segment, phase III (exists only if $q_i < q_{\text{nr}}$), with $a = \max(L_{\min}, L_i, L_j)$ and $b = \min(L_{\max}, L_{\text{nr}})$. In the same way,

$$\begin{aligned} \int_a^b \phi(L) R_{\geq i}(L) dL = \frac{1}{2} \theta_j^2 R_{\text{int}} \frac{\alpha + 1}{L_{\max}^{\alpha+1} - L_{\min}^{\alpha+1}} & \left[\frac{b^{\alpha+1} - a^{\alpha+1}}{\alpha + 1} \right. \\ & \left. + 2 L_j^{-\frac{1}{2p}} \frac{b^{\alpha+1+\frac{1}{2p}} - a^{\alpha+1+\frac{1}{2p}}}{\alpha + 1 + \frac{1}{2p}} + L_j^{-\frac{1}{p}} \frac{b^{\alpha+1+\frac{1}{p}} - a^{\alpha+1+\frac{1}{p}}}{\alpha + 1 + \frac{1}{p}} \right]. \quad (31) \end{aligned}$$

Saturated segment, with $a = \max(L_{\min}, L_i, L_{\text{nr}})$ and $b = L_{\max}$. Here $R_{\geq i} = R_{\text{int}}$ and the contribution is R_{int} times the fraction of the population in the segment:

$$\int_a^b \phi(L) R_{\geq i}(L) dL = R_{\text{int}} \frac{b^{\alpha+1} - a^{\alpha+1}}{L_{\max}^{\alpha+1} - L_{\min}^{\alpha+1}}. \quad (32)$$

The full result is the sum of (28)–(32). (The prefactor $\frac{1}{2} \theta_j^2 q_{\text{nr}}^2 = 1$ makes (32) consistent with the other three.) We now write it out for the configurations that occur in practice.

4.3 Assembled results

Case A: all bursts cadence-limited, $L_{\max} \leq L_i$. This is the situation of a survey whose flux limit is high compared with the population (range I–II of the thesis for every burst). Only (28) contributes, with $a = L_{\min}$, $b = L_{\max}$:

$$\begin{aligned} \langle R_{\geq i} \rangle &= \frac{1}{2} \theta_j^2 \min(q_i, q_{\text{nr}})^2 R_{\text{int}} L_i^{-3/2} \frac{\alpha + 1}{\alpha + 5/2} \frac{L_{\max}^{\alpha+5/2} - L_{\min}^{\alpha+5/2}}{L_{\max}^{\alpha+1} - L_{\min}^{\alpha+1}} \\ &= R_{\geq i}(L_{\text{eff}}), \quad \text{with} \quad L_{\text{eff}}^{3/2} \equiv \int_{L_{\min}}^{L_{\max}} \phi(L) L^{3/2} dL. \end{aligned} \quad (33)$$

The single-luminosity formula survives with L replaced by the effective luminosity $L_{\text{eff}} = \langle L^{3/2} \rangle^{2/3}$, the volume-weighted mean. For a broad distribution, $L_{\max} \gg L_{\min}$, its three limits are

$$\langle R_{\geq i} \rangle \approx \begin{cases} \frac{\alpha + 1}{\alpha + 5/2} R_{\geq i}(L_{\max}), & \alpha > -1, \\ \frac{|\alpha + 1|}{\alpha + 5/2} R_{\geq i}(L_{\max}) \left(\frac{L_{\min}}{L_{\max}} \right)^{|\alpha+1|} = \frac{|\alpha + 1|}{\alpha + 5/2} R_{\geq i}(L_{\min}) \left(\frac{L_{\max}}{L_{\min}} \right)^{\alpha+5/2}, & -\frac{5}{2} < \alpha < -1, \\ \frac{|\alpha + 1|}{|\alpha + 5/2|} R_{\geq i}(L_{\min}), & \alpha < -\frac{5}{2}. \end{cases} \quad (34)$$

The bright end dominates for $\alpha > -1$ and the faint end for $\alpha < -5/2$. In the middle range both ends matter: the rate is $R_{\geq i}(L_{\max})$ suppressed by the small fraction of bursts that are bright, $\sim (L_{\min}/L_{\max})^{|\alpha+1|}$, or equivalently $R_{\geq i}(L_{\min})$ boosted by the volume factor of the bright tail, $(L_{\max}/L_{\min})^{\alpha+5/2}$.

Case B: the distribution straddles L_i , no saturation, $L_{\min} < L_i < L_{\max} \leq L_{\text{nr}}$. This is the generic situation for $i \geq 2$ with a daily cadence: the fiducial burst sits near L_i , and the population extends to both sides. For $q_j < q_i < q_{\text{nr}}$ (the cadence angle beyond the jet-break angle, as for $i \geq 2$ and $t_{\text{cad}} \gtrsim 1$ d in Fig. 1) the phase-II segment is absent and (28) + (31) give

$$\begin{aligned} \langle R_{\geq i} \rangle &= \frac{1}{2} \theta_j^2 R_{\text{int}} \frac{\alpha + 1}{L_{\max}^{\alpha+1} - L_{\min}^{\alpha+1}} \left[q_i^2 L_i^{-3/2} \frac{L_i^{\alpha+5/2} - L_{\min}^{\alpha+5/2}}{\alpha + 5/2} + \frac{L_{\max}^{\alpha+1} - L_i^{\alpha+1}}{\alpha + 1} \right. \\ &\quad \left. + 2 L_j^{-\frac{1}{2p}} \frac{L_{\max}^{\alpha+1+\frac{1}{2p}} - L_i^{\alpha+1+\frac{1}{2p}}}{\alpha + 1 + \frac{1}{2p}} + L_j^{-\frac{1}{p}} \frac{L_{\max}^{\alpha+1+\frac{1}{p}} - L_i^{\alpha+1+\frac{1}{p}}}{\alpha + 1 + \frac{1}{p}} \right], \end{aligned} \quad (35)$$

where $q_i^2 = (1 + (L_i/L_j)^{1/(2p)})^2$ by (11), so that the cadence-limited and Euclidean-limited parts join continuously at L_i . The first term collects the bursts fainter than L_i , whose horizon lies inside the volume; the remaining three collect the brighter bursts, which are seen throughout the volume within the widening cone $q_{\text{Euc}}(L)$. If instead $q_{\text{dec}} < q_i < q_j$ (short cadence or small i , so that $L_i < L_j$), the phase-II segment (30) over $[L_i, \min(L_{\max}, L_j)]$ is inserted between the two, and the phase-III segment starts at L_j .

Case C: the bright end saturates, $L_{\max} > L_{\text{nr}}$. The phase-III segment is cut at L_{nr} and the saturated segment (32) is added; for $L_{\min} < L_i$ and $q_j < q_i < q_{\text{nr}}$,

$$\begin{aligned} \langle R_{\geq i} \rangle &= \frac{1}{2} \theta_j^2 R_{\text{int}} \frac{\alpha + 1}{L_{\max}^{\alpha+1} - L_{\min}^{\alpha+1}} \left[q_i^2 L_i^{-3/2} \frac{L_i^{\alpha+5/2} - L_{\min}^{\alpha+5/2}}{\alpha + 5/2} + \frac{L_{\text{nr}}^{\alpha+1} - L_i^{\alpha+1}}{\alpha + 1} \right. \\ &\quad \left. + 2 L_j^{-\frac{1}{2p}} \frac{L_{\text{nr}}^{\alpha+1+\frac{1}{2p}} - L_i^{\alpha+1+\frac{1}{2p}}}{\alpha + 1 + \frac{1}{2p}} + L_j^{-\frac{1}{p}} \frac{L_{\text{nr}}^{\alpha+1+\frac{1}{p}} - L_i^{\alpha+1+\frac{1}{p}}}{\alpha + 1 + \frac{1}{p}} \right] \\ &\quad + R_{\text{int}} \frac{L_{\max}^{\alpha+1} - L_{\text{nr}}^{\alpha+1}}{L_{\max}^{\alpha+1} - L_{\min}^{\alpha+1}}. \end{aligned} \quad (36)$$

The last term is the fraction of the population brighter than L_{nr} , each of which is detected with certainty. For $\alpha > -1$ and $L_{\text{max}} \gg L_{\text{nr}}$ this fraction tends to one and $\langle R_{\geq i} \rangle \rightarrow R_{\text{int}}$; for $\alpha < -1$ it is a small number $\approx (L_{\text{min}}/L_{\text{nr}})^{|\alpha+1|}$.

Case D: the single-luminosity limit, $L_{\text{max}} \rightarrow L_{\text{min}}$. Then $\phi \rightarrow \delta(L - L_{\text{min}})$ and (26) gives $R_{\geq i}(L_{\text{min}})$. Explicitly, in (33), $\frac{\alpha+1}{\alpha+5/2} \frac{L_{\text{max}}^{\alpha+5/2} - L_{\text{min}}^{\alpha+5/2}}{L_{\text{max}}^{\alpha+1} - L_{\text{min}}^{\alpha+1}} \rightarrow L_{\text{min}}^{3/2}$ by l'Hôpital's rule, and likewise for every other segment: the results of the thesis are recovered exactly.

4.4 Power-law description and which luminosities dominate

Following thesis Eqs. 53 and 77, we drop the “+1” in the Euclidean-limited lines of (18) where $\tilde{q}_{\text{Euc}} \gg 1$ (phase III) and keep only the “1” where $\tilde{q}_{\text{Euc}} \ll 1$ (phase II), so that each line is a pure power of L :

$$R_{\geq i}(L) \approx \frac{1}{2} \theta_j^2 R_{\text{int}} \times \begin{cases} q_i^2 (L/L_i)^{3/2}, & L < L_i, \\ 1, & L_i < L < L_j \text{ (if } q_i < q_j), \\ (L/L_j)^{1/p}, & \max(L_i, L_j) < L < L_{\text{nr}}, \\ 2\theta_j^{-2}, & L > L_{\text{nr}}. \end{cases} \quad (37)$$

The integrand of (26) is then $\phi(L)R_{\geq i}(L) \propto L^{\alpha+3/2}$ below L_i , $\propto L^\alpha$ or $L^{\alpha+1/p}$ in the Euclidean-limited range, and $\propto L^\alpha$ once saturated. Reading off where each power rises or falls gives three regimes (Fig. 2):

index	integrand $\phi(L)R_{\geq i}(L)$	detections dominated by
$\alpha < -\frac{5}{2}$	falls everywhere	the faint end L_{min} : many faint bursts seen in a small volume
$-\frac{5}{2} < \alpha < -1$	rises below L_i , falls above	the knee L_i : bursts whose cadence horizon just reaches D_{Euc}
$\alpha > -1$	rises everywhere (until saturation)	the bright end L_{max} : few bright bursts seen throughout the volume

(For $-1-1/p < \alpha < -1$ the phase-III integrand $\propto L^{\alpha+1/p}$ rises slowly, so the bright end competes with the knee unless L_{max} is close to L_i ; the boundary between the last two regimes is therefore soft.) The three regimes have different leading-order results, which are the analytic content of the luminosity distribution:

Faint-end dominated, $\alpha < -5/2$: from (34),

$$\langle R_{\geq i} \rangle \approx \frac{|\alpha+1|}{|\alpha+5/2|} R_{\geq i}(L_{\text{min}}) \quad (L_{\text{min}} \ll L_i), \quad (38)$$

the single-luminosity rate at L_{min} with an order-unity factor; all scalings with F_{lim} and t_{cad} are those of thesis Eq. 77.

Knee dominated, $-5/2 < \alpha < -1$ and $L_{\text{min}} \ll L_i \ll L_{\text{max}}$: keeping the leading term of each bracket in (35) (the $L_i^{\alpha+5/2}$ term of the first and the $L_i^{\alpha+1}$ terms of the others, both of which are the values at the knee),

$$\langle R_{\geq i} \rangle \approx \frac{1}{2} \theta_j^2 q_i^2 R_{\text{int}} \frac{(\alpha+1) L_i^{\alpha+1}}{L_{\text{max}}^{\alpha+1} - L_{\text{min}}^{\alpha+1}} \left[\frac{1}{\alpha+5/2} + \frac{1}{|\alpha+1|} \right] \approx R_{\geq i}(L_i) \left(1 + \frac{|\alpha+1|}{\alpha+5/2} \right) \left(\frac{L_{\text{min}}}{L_i} \right)^{|\alpha+1|}, \quad (39)$$

where the second form uses $L_{\text{max}}^{\alpha+1} \ll L_{\text{min}}^{\alpha+1}$. For $\alpha = -2$ this is $\langle R_{\geq i} \rangle \approx 3 R_{\geq i}(L_i) L_{\text{min}}/L_i$: the rate is that of a burst at the knee, $R_{\geq i}(L_i) = \frac{1}{2} \theta_j^2 q_i^2 R_{\text{int}}$ (which is independent of F_{lim}), times

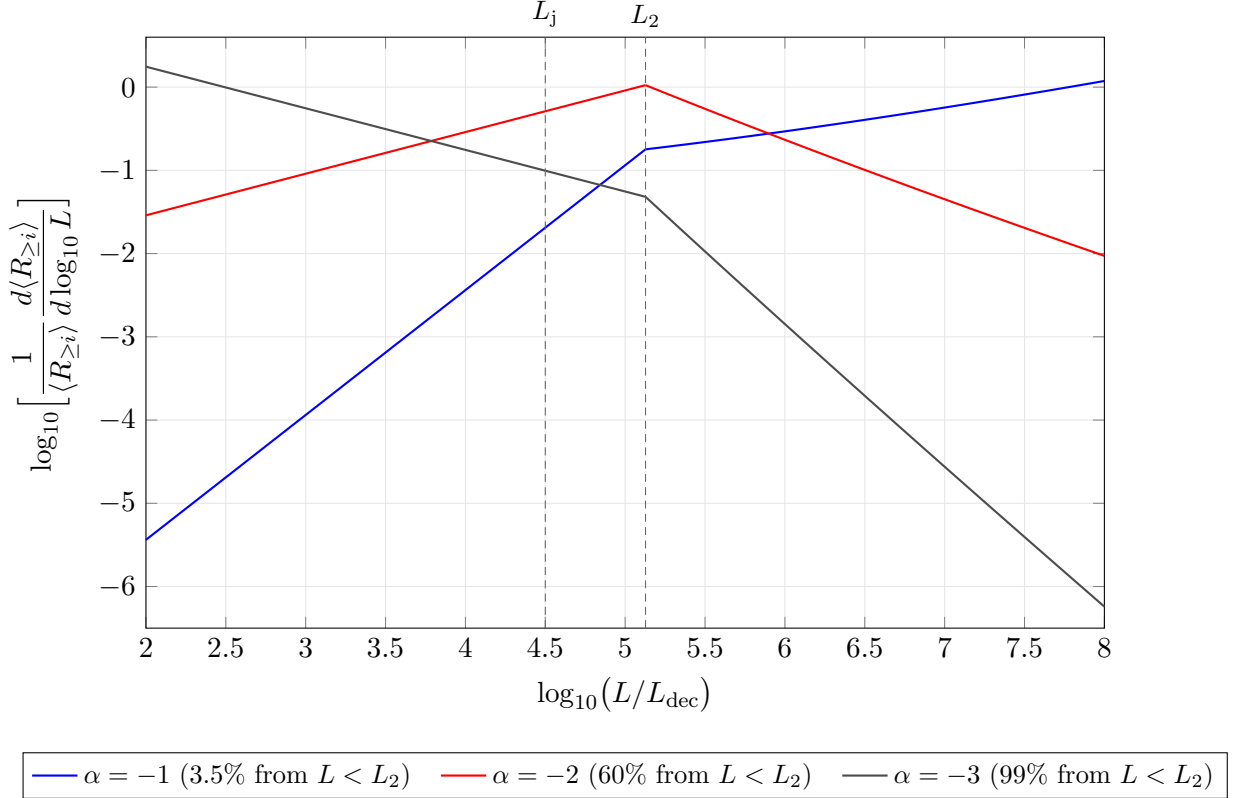


Figure 2: Where the detections come from: the detected-luminosity density per decade, normalised to the total rate, for $i = 2$, $t_{\text{cad}} = 2$ d, the thesis parameters, and a population spanning $L_{\text{min}} = 10^2 L_{\text{dec}}$ to $L_{\text{max}} = 10^8 L_{\text{dec}}$ with three values of α . The slope is $\alpha + 5/2$ below the knee L_2 and $\alpha + 1 + 1/p$ above it. For $\alpha = -3$ the faint end dominates, for $\alpha = -2$ the bursts near the knee, for $\alpha = -1$ the bright end. The percentages are the fractions of $\langle R_{\geq i} \rangle$ contributed by bursts fainter than L_2 (from Eqs. (28) and (31)); the normalisations are listed in Appendix B.

the fraction of the population within a factor of a few of the knee. In the cases of Fig. 2 the approximation (39) is within 10% of the full expression (35).

The consequences for the survey scalings are the most important result of this section. Since $R_{\geq i}(L_i)$ does not depend on F_{lim} and $L_i \propto F_{\text{lim}}$ (Eq. (15)),

$$\langle R_{\geq i} \rangle \propto q_i^2 L_i^{\alpha+1} \propto F_{\text{lim}}^{\alpha+1} \left(\frac{i t_{\text{cad}}}{t_j} \right)^{1+p(\alpha+1)} \quad (i t_{\text{cad}} > t_j, \text{ knee dominated}), \quad (40)$$

to be compared with the single-luminosity $R_{\geq i} \propto F_{\text{lim}}^{-3/2} (i t_{\text{cad}}/t_j)^{-3(p-2)/2}$ of thesis Eq. 77. For $\alpha = -2$ and $p = 2.5$ the dependence on the limiting flux softens from $F_{\text{lim}}^{-3/2}$ to F_{lim}^{-1} – lowering the flux limit by a factor ten gains a factor ten in detections instead of thirty – while the dependence on the cadence steepens from $(i t_{\text{cad}})^{-0.75}$ to $(i t_{\text{cad}})^{-1.5}$. The reason is that a deeper limit no longer enlarges the volume in which a fixed burst is seen; it lowers the knee L_i , admitting the more numerous fainter bursts at the rate $\phi(L_i)L_i \propto L_i^{\alpha+1}$.

Bright-end dominated, $\alpha > -1$: for $L_{\text{max}} < L_{\text{nr}}$ the last bracket of (35) dominates and

$$\langle R_{\geq i} \rangle \approx \frac{\alpha + 1}{\alpha + 1 + 1/p} R_{\geq i}(L_{\text{max}}) \quad (L_i \ll L_{\text{max}} < L_{\text{nr}}), \quad (41)$$

while for $L_{\text{max}} \gg L_{\text{nr}}$ the saturated term of (36) takes over and $\langle R_{\geq i} \rangle \rightarrow R_{\text{int}}$: essentially every burst inside D_{Euc} is bright enough to be seen from anywhere inside the volume.

5 The luminosities of the detected bursts

The integrand of (26) is itself an observable: the distribution of luminosities among the *detected* bursts,

$$\frac{d\langle R_{\geq i} \rangle}{dL} = \phi(L) R_{\geq i}(L), \quad \text{or per decade} \quad \frac{d\langle R_{\geq i} \rangle}{d\log_{10} L} = \ln(10) L \phi(L) R_{\geq i}(L), \quad (42)$$

which is the intrinsic distribution re-weighted by the detection volume. Below the knee the weight is $L^{3/2}$: with $\alpha = -2$ the intrinsic population per decade falls as L^{-1} while the detected one *rises* as $L^{1/2}$ up to L_i (Fig. 2). The detected sample is therefore strongly biased towards luminous bursts (the Malmquist bias of a flux-limited survey), and the median luminosity of the detected bursts,

$$\int_{L_{\min}}^{L_{\text{med,det}}} \phi(L) R_{\geq i}(L) dL = \frac{1}{2} \langle R_{\geq i} \rangle, \quad (43)$$

lies far above the population median (5). In case A (all cadence-limited) it is explicit,

$$L_{\text{med,det}}^{\alpha+5/2} = \frac{1}{2} (L_{\min}^{\alpha+5/2} + L_{\max}^{\alpha+5/2}) \quad (L_{\max} \leq L_i), \quad (44)$$

which for $\alpha = -2$ and $L_{\max} \gg L_{\min}$ gives $L_{\text{med,det}} \approx L_{\max}/4$, against $L_{\text{med,pop}} = 2L_{\min}$. In the general case the median is found segment by segment from (28)–(32); for the $\alpha = -2$ example of Fig. 2 it is $L_{\text{med,det}} \approx 10^{5.0} L_{\text{dec}}$, three orders of magnitude above $L_{\text{med,pop}} = 10^{2.3} L_{\text{dec}}$, and just below the knee $L_2 = 10^{5.1} L_{\text{dec}}$.

Conversely, the *effective luminosity* at which the single-luminosity model of the thesis reproduces the population rate, $R_{\geq i}(L_{\text{eff}}) = \langle R_{\geq i} \rangle$, is obtained by inverting (18). In case A it is the volume-weighted mean $L_{\text{eff}} = \langle L^{3/2} \rangle^{2/3}$ of (33); in the knee-dominated regime, where $\langle R_{\geq i} \rangle < R_{\geq i}(L_i)$, it lies on the rising branch and (39) gives

$$L_{\text{eff}} \approx L_i \left[\left(1 + \frac{|\alpha + 1|}{\alpha + 5/2} \right) \left(\frac{L_{\min}}{L_i} \right)^{|\alpha+1|} \right]^{2/3} \quad (-\frac{5}{2} < \alpha < -1), \quad (45)$$

e.g. $L_{\text{eff}} \approx L_i (3L_{\min}/L_i)^{2/3}$ for $\alpha = -2$ (for the example of Fig. 2, $L_{\text{eff}} \approx 10^{3.4} L_{\text{dec}}$). Note that L_{eff} depends on the survey through $L_i \propto F_{\text{lim}}(i t_{\text{cad}})^p$: no single luminosity reproduces the population rate for all surveys, which is the quantitative statement of why a distribution is needed.

6 Summary of results

Model. Identical light-curve shapes; luminosities $L \equiv L_{\nu}(t_{\text{dec}})$ distributed as $\phi(L) \propto L^{\alpha}$ on $[L_{\min}, L_{\max}]$, normalised so that R_{int} is the total GRB rate inside D_{Euc} . Survey with limiting flux F_{lim} and cadence t_{cad} ; at least $i \geq 2$ detections required; $q_{\text{dec}} < q_i$; step approximation for $P_{\geq i}$; most-contributing-point approximation for the (q, D) integral.

Scalings. Luminosity enters only through $\tilde{F}_{\text{lim}} = L_{\text{dec}}/L$ with $L_{\text{dec}} = 4\pi D_{\text{Euc}}^2 F_{\text{lim}}$. All distances scale as $L^{1/2}$; all angles and times except $q_{\text{Euc}}(L)$ are fixed. The regime of a burst is set by its luminosity relative to the threshold luminosities $L_X = L_{\text{dec}}/\tilde{F}_p(q_X)$ at which $q_{\text{Euc}}(L_X) = q_X$:

$$L_{\text{dec}} = 4\pi D_{\text{Euc}}^2 F_{\text{lim}}, \quad L_j = L_{\text{dec}} (\Gamma_0 \theta_j)^{2(p-1)}, \quad L_{\text{nr}} = L_j \tilde{q}_{\text{nr}}^{2p},$$

$$L_i = \frac{L_{\text{dec}}}{\tilde{F}_{\nu}(i t_{\text{cad}})} = L_j \left(\frac{i t_{\text{cad}}}{t_j} \right)^p \quad (i t_{\text{cad}} > t_j).$$

Single-burst rate (Eq. (18)):

$$R_{\geq i}(L) = \frac{1}{2} \theta_j^2 R_{\text{int}} \times \begin{cases} \min(q_i, q_{\text{nr}})^2 (L/L_i)^{3/2}, & L < L_i, \\ (1 + (L/L_j)^{1/(2(p-1))})^2, & L_i < L < L_j, \\ (1 + (L/L_j)^{1/(2p)})^2, & \max(L_i, L_j) < L < L_{\text{nr}}, \\ 2\theta_j^{-2}, & L > \max(L_i, L_{\text{nr}}). \end{cases}$$

Population rate (Eq. (26)): $\langle R_{\geq i} \rangle = \int \phi(L) R_{\geq i}(L) dL$, evaluated in closed form as the sum of the segment contributions (28)–(32); assembled results for the practical cases in Eqs. (33), (35), (36). The exact (non-approximated) rate is the single q integral (25) with the closed-form luminosity integral (24).

Which bursts are detected. $\alpha < -5/2$: the faint end, $\langle R_{\geq i} \rangle \approx \frac{|\alpha+1|}{|\alpha+5/2|} R_{\geq i}(L_{\min})$. $-5/2 < \alpha < -1$: the knee, $\langle R_{\geq i} \rangle \approx R_{\geq i}(L_i) (1 + \frac{|\alpha+1|}{\alpha+5/2}) (L_{\min}/L_i)^{|\alpha+1|} \propto F_{\lim}^{\alpha+1} (i t_{\text{cad}}/t_j)^{1+p(\alpha+1)}$. $\alpha > -1$: the bright end, $\langle R_{\geq i} \rangle \approx \frac{\alpha+1}{\alpha+1+1/p} R_{\geq i}(L_{\max})$, saturating at R_{int} once $L_{\max} \gg L_{\text{nr}}$. The detected luminosities follow $\phi(L) R_{\geq i}(L)$ and are strongly biased towards the knee and above.

Limitations. The gray assumption ignores the co-variation of t_{dec} and t_j with the energy; the step approximation for $P_{\geq i}$ and the most-contributing-point approximation carry the same few-tens-of-per-cent accuracy as in the thesis; the case $q_i < q_{\text{dec}}$ ($i t_{\text{cad}} < t_{\text{dec}}$) is excluded; and, as in the thesis, the universe is Euclidean with a sharp edge at D_{Euc} .

A The logarithmic special cases

Whenever an exponent in (27) equals -1 the power bracket is replaced by a logarithm; each such case is the continuous limit of the general formula and no result changes discontinuously. The values of α concerned and the replacements are:

value of α	where it appears	replacement
$\alpha = -1$	normalisation of ϕ ; “1” terms of (30), (31); saturated segment (32)	$\frac{b^{\alpha+1} - a^{\alpha+1}}{\alpha + 1} \rightarrow \ln \frac{b}{a}$
$\alpha = -\frac{5}{2}$	cadence-limited segment (28)	$\frac{b^{\alpha+5/2} - a^{\alpha+5/2}}{\alpha + 5/2} \rightarrow \ln \frac{b}{a}$
$\alpha = -1 - \frac{1}{2(p-1)}, -1 - \frac{1}{p-1}$	cross and square terms of the phase-II segment (30)	the corresponding bracket $\rightarrow \ln \frac{b}{a}$
$\alpha = -1 - \frac{1}{2p}, -1 - \frac{1}{p}$	cross and square terms of the phase-III segment (31)	the corresponding bracket $\rightarrow \ln \frac{b}{a}$

For example, at $\alpha = -1$ the distribution is $\phi(L) = 1/(L \ln(L_{\max}/L_{\min}))$ and case A becomes

$$\langle R_{\geq i} \rangle = \frac{1}{2} \theta_j^2 \min(q_i, q_{\text{nr}})^2 R_{\text{int}} L_i^{-3/2} \frac{2}{3} \frac{L_{\max}^{3/2} - L_{\min}^{3/2}}{\ln(L_{\max}/L_{\min})} \quad (\alpha = -1),$$

while at $\alpha = -5/2$ it becomes

$$\langle R_{\geq i} \rangle = \frac{1}{2} \theta_j^2 \min(q_i, q_{\text{nr}})^2 R_{\text{int}} L_i^{-3/2} \frac{3}{2} \frac{\ln(L_{\max}/L_{\min})}{L_{\min}^{-3/2} - L_{\max}^{-3/2}} \quad (\alpha = -5/2).$$

B Parameters used in the figures

Both figures use the thesis parameters (thesis footnote 4): $p = 2.5$, $\Gamma_0 = 10^{2.5}$, $\theta_j = 0.1$ rad, $E_{\text{k,iso}} = 10^{53}$ erg, $n = 1 \text{ cm}^{-3}$, giving $t_{\text{dec}} = 19.4$ s, $\tilde{t}_j = 10^4$ ($t_j = 2.24$ d), $\tilde{q}_{\text{dec}} = (\Gamma_0 \theta_j)^{-1} = 10^{-1.5}$, $q_{\text{dec}} = 1.032$, $q_{\text{nr}} = 14.14$, $\tilde{q}_{\text{nr}} = 13.14$, and $\frac{1}{2} \theta_j^2 = 10^{-2.30}$. The threshold luminosities in units of L_{dec} are then, from (13)–(15),

$$\frac{L_j}{L_{\text{dec}}} = (\Gamma_0 \theta_j)^{2(p-1)} = 10^{4.50}, \quad \frac{L_{\text{nr}}}{L_{\text{dec}}} = \frac{L_j}{L_{\text{dec}}} \tilde{q}_{\text{nr}}^{2p} = 10^{10.09},$$

(reproducing $\tilde{F}_{\text{lim},j} = 10^{-4.5}$ and $\tilde{F}_{\text{lim},\text{nr}} = 10^{-10.1}$ of thesis Fig. 11), and for $t_{\text{cad}} = 2$ d

$$\begin{aligned} i = 2 : \quad & \frac{i t_{\text{cad}}}{t_j} = 1.78, \quad q_2 = 2.335, \quad \frac{L_2}{L_{\text{dec}}} = 10^{5.13}; \\ i = 10 : \quad & \frac{i t_{\text{cad}}}{t_j} = 8.91, \quad q_{10} = 3.985, \quad \frac{L_{10}}{L_{\text{dec}}} = 10^{6.87}. \end{aligned}$$

(With $t_{\text{cad}} = 1$ d and $i = 2$ one has $i t_{\text{cad}} < t_j$, $q_2 = 1.958$ just inside phase II, and $L_2 = 10^{4.44} L_{\text{dec}}$ practically coincides with L_j ; the two-day cadence was chosen to separate the two thresholds in the figure.) In Fig. 2 the population spans $[L_{\text{min}}, L_{\text{max}}] = [10^2, 10^8] L_{\text{dec}}$; the total rates from Eq. (35) are $\langle R_{\geq i} \rangle / R_{\text{int}} = 2.54 \times 10^{-2}$, 4.43×10^{-5} and 1.45×10^{-6} for $\alpha = -1, -2, -3$, to be compared with $R_{\geq 2}(L_2) / R_{\text{int}} = 1.82 \times 10^{-2}$ for a single burst at the knee; the fractions contributed by $L < L_2$ are 3.5%, 59.5% and 99.2%; the detected medians are $10^{6.9}$, $10^{5.0}$, $10^{2.5} L_{\text{dec}}$ against population medians of $10^{5.0}$, $10^{2.3}$, $10^{2.15} L_{\text{dec}}$; and the knee approximation (39) for $\alpha = -2$ is 0.92 of the full result. All closed forms of Section 4 were checked against direct numerical integration of (26) and Eq. (25) against numerical integration of (21).